East West University
Department of Computer Science and Engineering
Course: CSE464
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Lab 01: Introduction to PL/SQL - Anonymous Blocks & Control Structures

Lab Objectives

• Understand PL/SQL blocks (DECLARE, BEGIN, EXCEPTION, END)
• Learn how to use variables and constants
• Read user input using & substitution variable

Topics Covered

• PL/SQL Block Structure
• Variables and Constants
• Reading User Input

Lab Outcomes

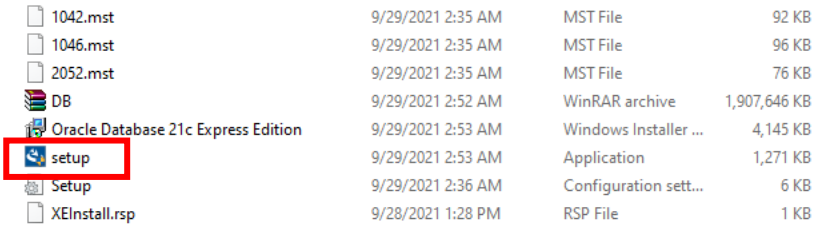
After completing this lab successfully, students will be able to:
1. Understand PL/SQL Syntax – Write and execute basic PL/SQL blocks.
2. Use Variables and Constants – Declare and manipulate variables and constants in PL/SQL.
3. Implement Loops – Develop iterative solutions using FOR, WHILE, and LOOP constructs.
4. Apply Conditional Statements – Implement IF-THEN-ELSE and CASE statements for decision-making.
5. Read User Input – Use & substitution variables to take input.
6. Insert and Retrieve Data – Create a table, insert some values into it, and interact with the table using SELECT queries.

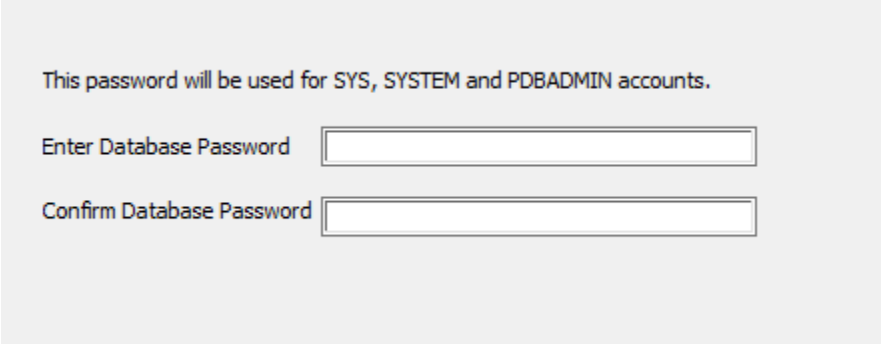
Lab Activities

1. Steps to Download and Set Up Oracle Database Express Edition (XE) and SQL*Plus for PL/SQL

    - **Search:**
      Open your browser and search for:
      "Oracle Database Express Edition (XE) Downloads"

- **Download:**
  Go to the official Oracle website and download the version suitable for your operating system (e.g., Windows, Linux).
- **Unzip:**
  After downloading, **unzip the installation package** to a preferred location on your computer.





2. Introduction to PL/SQL

PL/SQL (Procedural Language/Structured Query Language) is Oracle Corporation's procedural extension to SQL. It combines the data manipulation power of SQL with the processing capabilities of procedural programming languages. PL/SQL allows developers to write complex database operations using control structures such as loops, conditions, and exception handling.

It is primarily used to create procedures, functions, triggers, and cursors in Oracle databases.

Key Features:

- Integration of SQL and procedural logic
- Support for variables, conditions, and loops
- Exception handling for runtime errors
- Block-structured programming for modularity

- High performance in executing database operations

PL/SQL Block Structure
A PL/SQL program is written as a **block**, which has three main sections:

1. **Declaration Section** – for declaring variables, constants, cursors, etc. (optional)
2. **Execution Section** – where the actual code logic is written (mandatory)
3. **Exception Handling Section** – for handling runtime errors (optional)

Structure of a Block
DECLARE

/* Declarative section: variables, types, cursors, user-defined exceptions */

BEGIN

/* Executable section: procedural and SQL statements go here. */

/* this is the only section of the block that is required. */

EXCEPTION

/* Exception handling section: error handling statements go here. */
END; /* mandatory */

Note: Always end the block with / (forward slash) when running in tools like SQL*Plus to execute the block.

3. Write a PL/SQL program to print "Hello World".

Solution:

```
BEGIN
    DBMS_OUTPUT.PUT_LINE('Hello World');
END;
/
```

4. Write a PL/SQL program to print a number.

Solution:

```
DECLARE
    num NUMBER := 100;
BEGIN
    DBMS_OUTPUT.PUT_LINE('The number is: ' || num);
END;
/
```

5. Write a PL/SQL program to add two numbers (taking user input).

Solution:

```
DECLARE
    num1 NUMBER;
    num2 NUMBER;
    sum_result NUMBER;
BEGIN
    num1 := &Enter_First_Number;
    num2 := &Enter_Second_Number;

    sum_result := num1 + num2;

    DBMS_OUTPUT.PUT_LINE('The sum is: ' || sum_result);
END;
/
```

6. Write a PL/SQL program to print your name and roll number (taking user input).

Solution:

```
DECLARE
    my_name VARCHAR2(30);
    my_roll NUMBER;
BEGIN
    my_name := '&Enter_Your_Name';
    my_roll := &Enter_Your_Roll_Number;

    DBMS_OUTPUT.PUT_LINE('My Name is: ' || my_name);
    DBMS_OUTPUT.PUT_LINE('My Roll Number is: ' || my_roll);
END;
/
```

7. Create an `Employees` table with three attributes: `employee_id` (Primary Key), `name`, and `salary`. Insert some sample data into it.